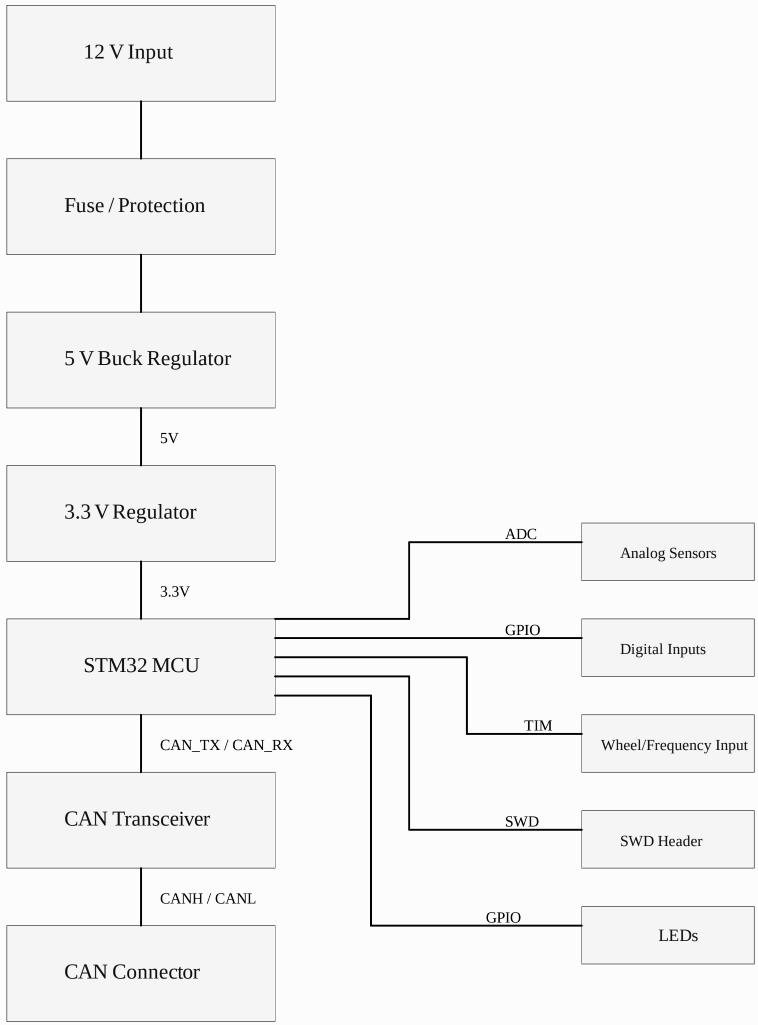
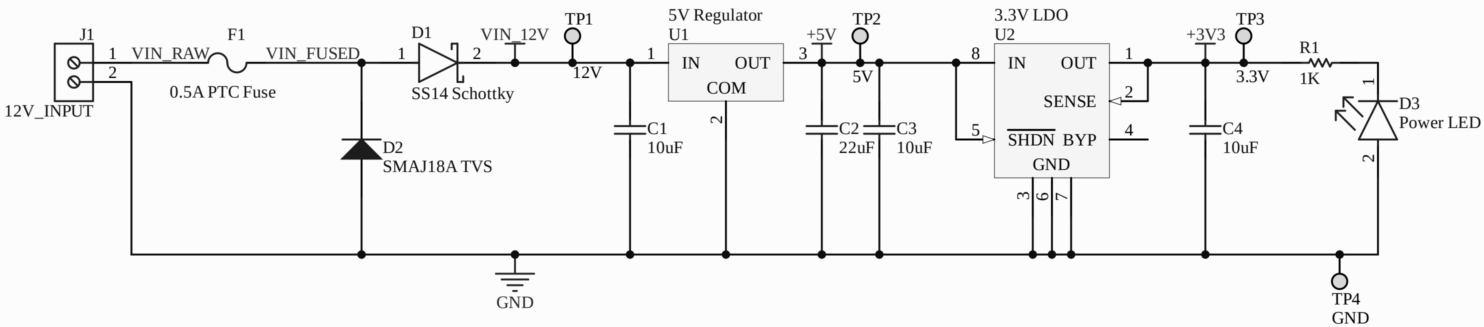
Purpose: Converts vehicle/sensor inputs into CAN telemetry using an STM32 microcontroller.



Title	STM32 CAN Telemetry PCB — Block Diagram		
Size A	Number 00		Revision A
Date:	6/19/2026	Sheet of	1 of 1
File:	C:\Users\..\00_Block_Diagram.SchDoc	Drawn By:	Joshua Nwaneli

Power architecture: 12 V vehicle input is protected, regulated to 5 V, then regulated to 3.3 V for STM32 logic.

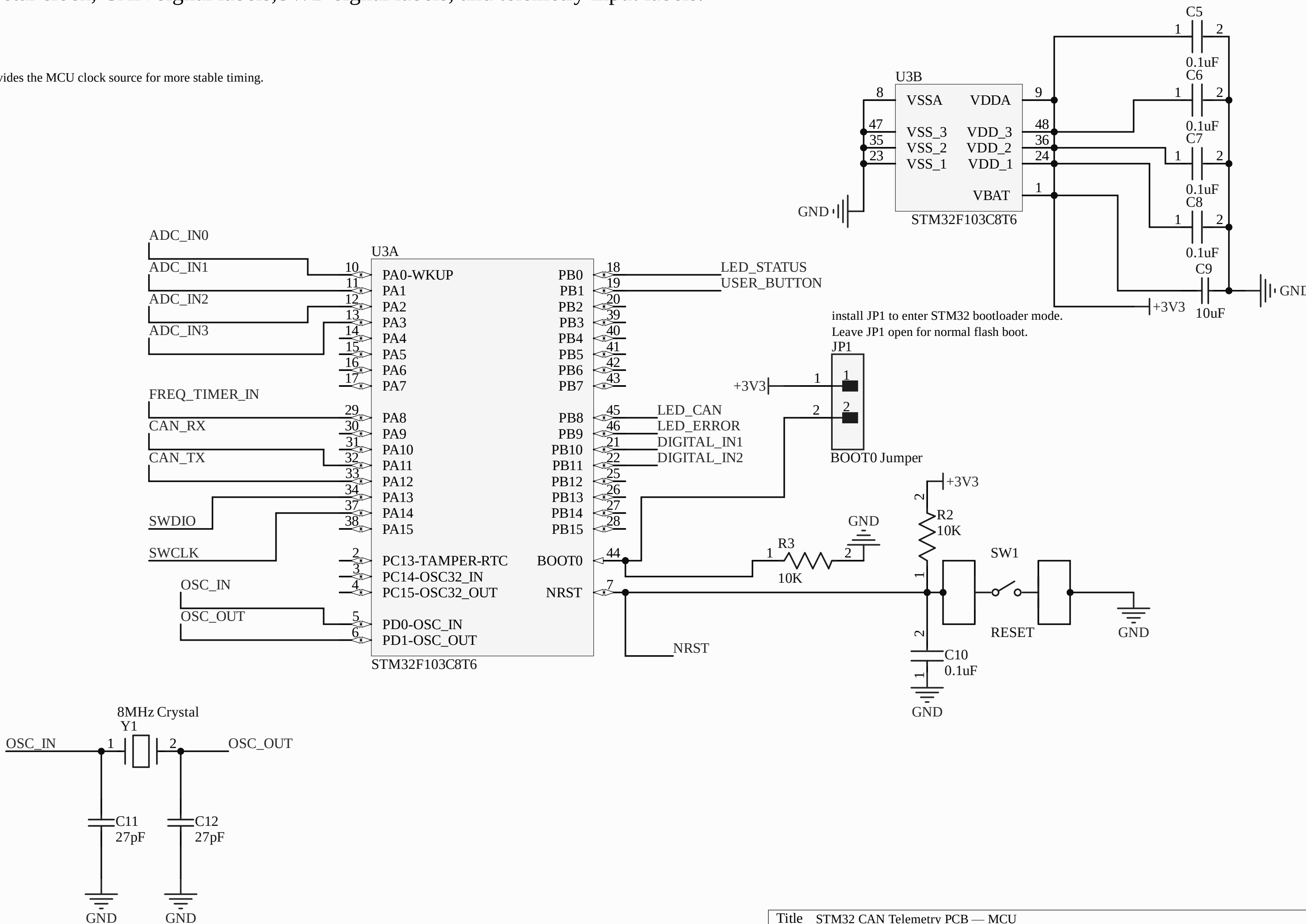
Do not power STM32 directly from +5V. STM32 VDD uses +3V3.



Title	STM32 CAN Telemetry PCB — Power Supply		
Size A	Number 01		Revision A
Date:	6/19/2026	Sheet of	1 of 1
File:	C:\Users\..\01_Power.SchDoc	Drawn By:	Joshua Nwaneli

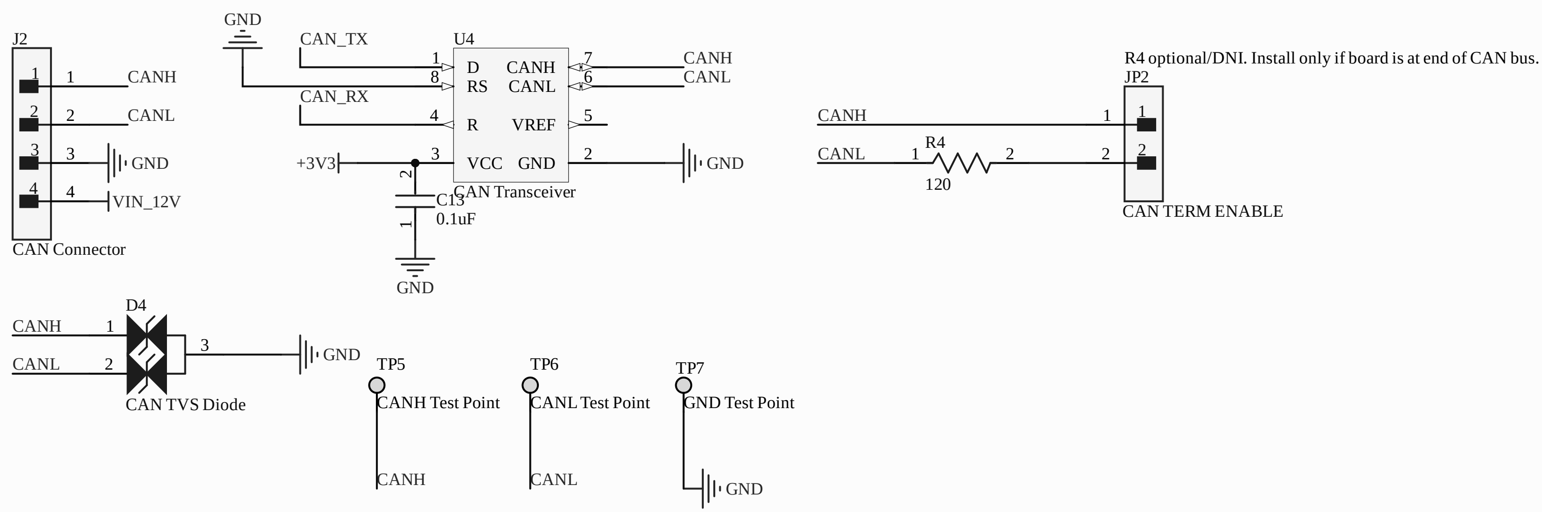
Purpose: Provides the STM32 controller, 3.3 V power connections, reset/boot circuitry external crystal clock, CAN signal labels,SWD signal labels, and telemetry input labels.

External crystal provides the MCU clock source for more stable timing.



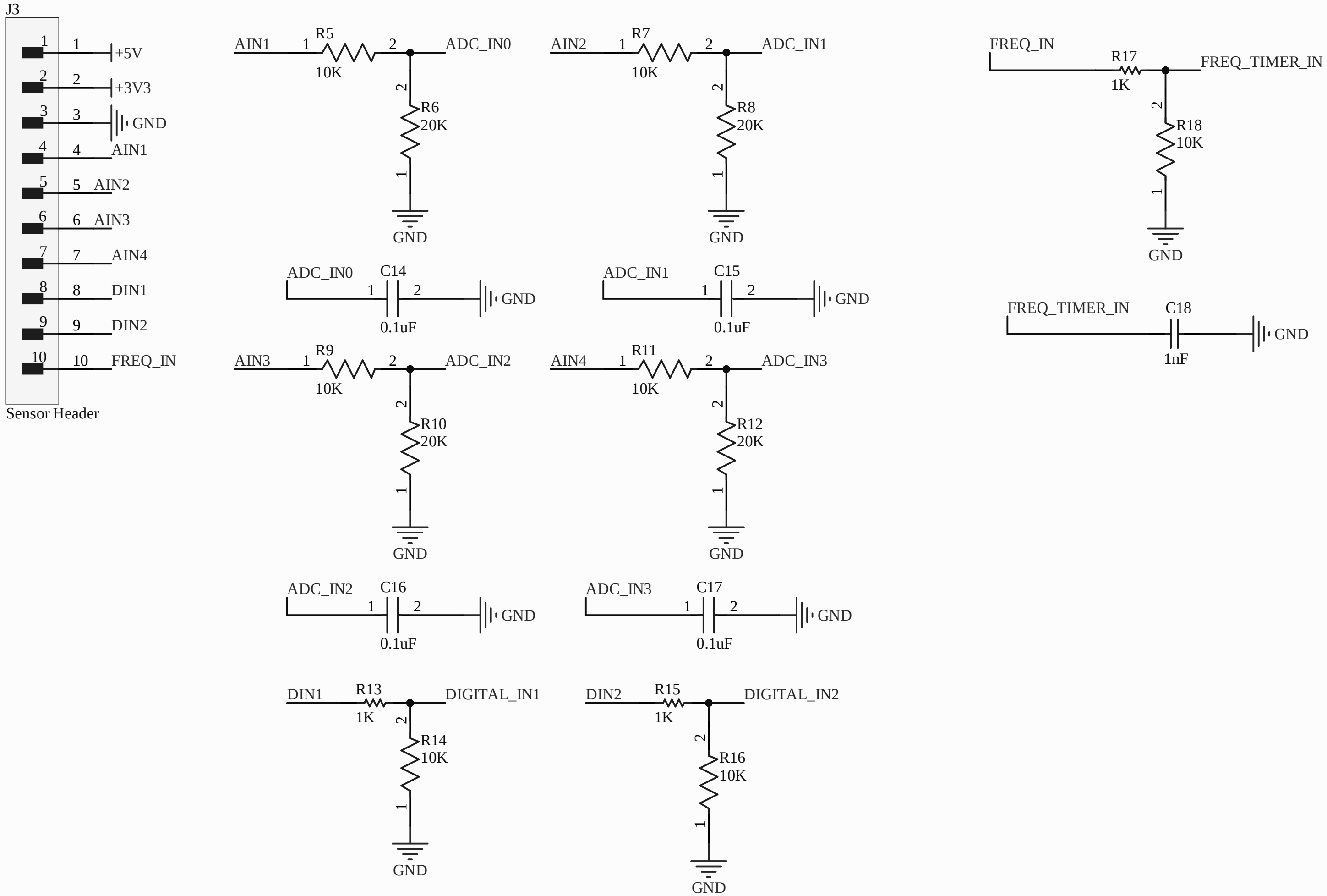
Title	STM32 CAN Telemetry PCB — MCU		
Size A	Number 02		Revision A
Date:	6/19/2026	Sheet of	1 of 1
File:	C:\Users\..\02_MCU.SchDoc	Drawn By:	Joshua Nwaneli

Purpose: Converts STM32 CAN_TX/CAN_RX logic signals into differential CANH/CANL bus signals with connector, protection, test points, and optional termination.



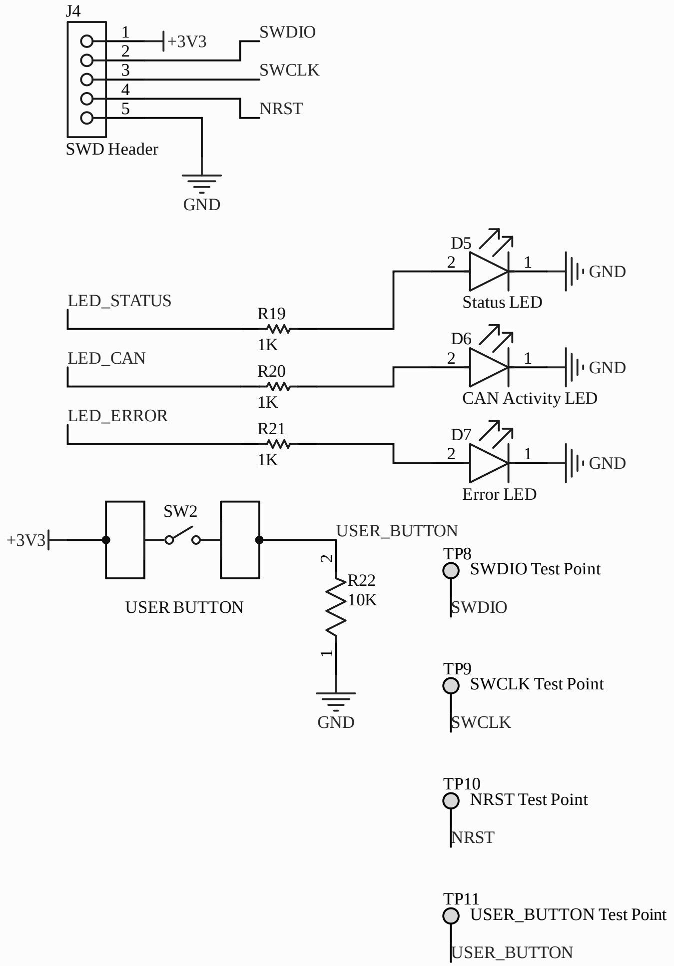
Title	STM32 CAN Telemetry PCB — CAN Interface		
Size A	Number 03		Revision A
Date:	6/19/2026	Sheet of	1 of 1
File:	C:\Users\..\03_CAN.SchDoc	Drawn By:	Joshua Nwaneli

Purpose: Provides analog, digital, and frequency input circuits for vehicle telemetry signals going into the STM32.



Title	STM32 CAN Telemetry PCB — Sensors and I/O		
Size A	Number 04		Revision A
Date:	6/19/2026	Sheet of	1 of 1
File:	C:\Users\..\04_Sensors_IO.SchDoc	Drawn By:	Joshua Nwaneli

Purpose: Provides SWD programming access, debug LEDs, user button, and optional signal test points for board bring-up and troubleshooting.



Title STM32 CAN Telemetry PCB — Connectors and Debug		
Size A	Number 05	Revision A
Date: 6/19/2026		Sheet of 1 of 1
File: C:\Users\..\05_Connectors_Debug.SchDoc		Drawn By: Joshua Nwaneli